VISUALIZING HOUSING MARKET TRENDS

AN ANALYSIS OF SALE PRICES AND FEATURES USING TABLEAU

Team Id: DataAnalytics10

Team Members:

1) Soniya Shinde
2) Sonali Mali
3) Anuj Nangare
4)Adinath Powar

1)INTRODUCTION:

1.1 Project Overview:

The project focuses on transforming ABC Company's raw housing market data into an interactive, Skill Wallet-style Tableau dashboard to help stakeholders understand factors that influence house prices and sales trends. The dashboard presents scenarios such as:

- **Scenario 1: Overall Data Overview** – Summary metrics including total records, average sale price, and total area.
- **Scenario 2: Total Sales by Years Since Renovation** – Histogram showing sales distribution based on years since renovation.
- **Scenario 3: Distribution of House Age by Renovation Status** – Pie chart illustrating age and renovation distribution.
- **Scenario 4: House Age Distribution by Number of Bathrooms, Bedrooms, and Floors** – Grouped bar chart detailing house features across age categories.

The project follows a **prototype-first approach**: stakeholders receive initial dashboards quickly to validate direction, allowing rapid iterations and alignment with business questions.

The dashboards are designed for scalability and self-service. Tableau Prep flows automate data refreshes, and Flask integration allows the dashboards to be embedded within web applications, enabling the team to explore trends dynamically.

1.2 Purpose:

- Transform raw housing market data into **interactive visualizations** that reveal price trends, house features, and sales patterns.

- Enable stakeholders, including analysts, marketing teams, and executives, to **make data-driven decisions** quickly.

- Provide a **scalable, repeatable workflow** so new datasets or regions can be incorporated with minimal effort.

- Highlight key insights, such as the impact of renovations, house age, and property features, to **optimize pricing strategies and market competitiveness**.

- Support **stakeholder collaboration** by providing dashboards that can be easily shared and interpreted.

- Enhance **forecasting and trend analysis**, helping the company anticipate market shifts and plan strategies proactively.

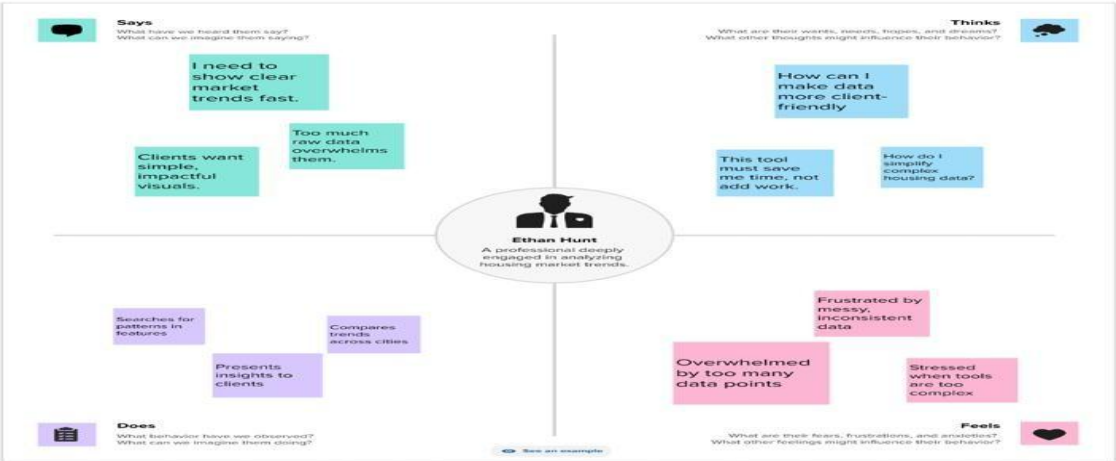
2)IDEATION PHASE:

2.1 Problem Statement:

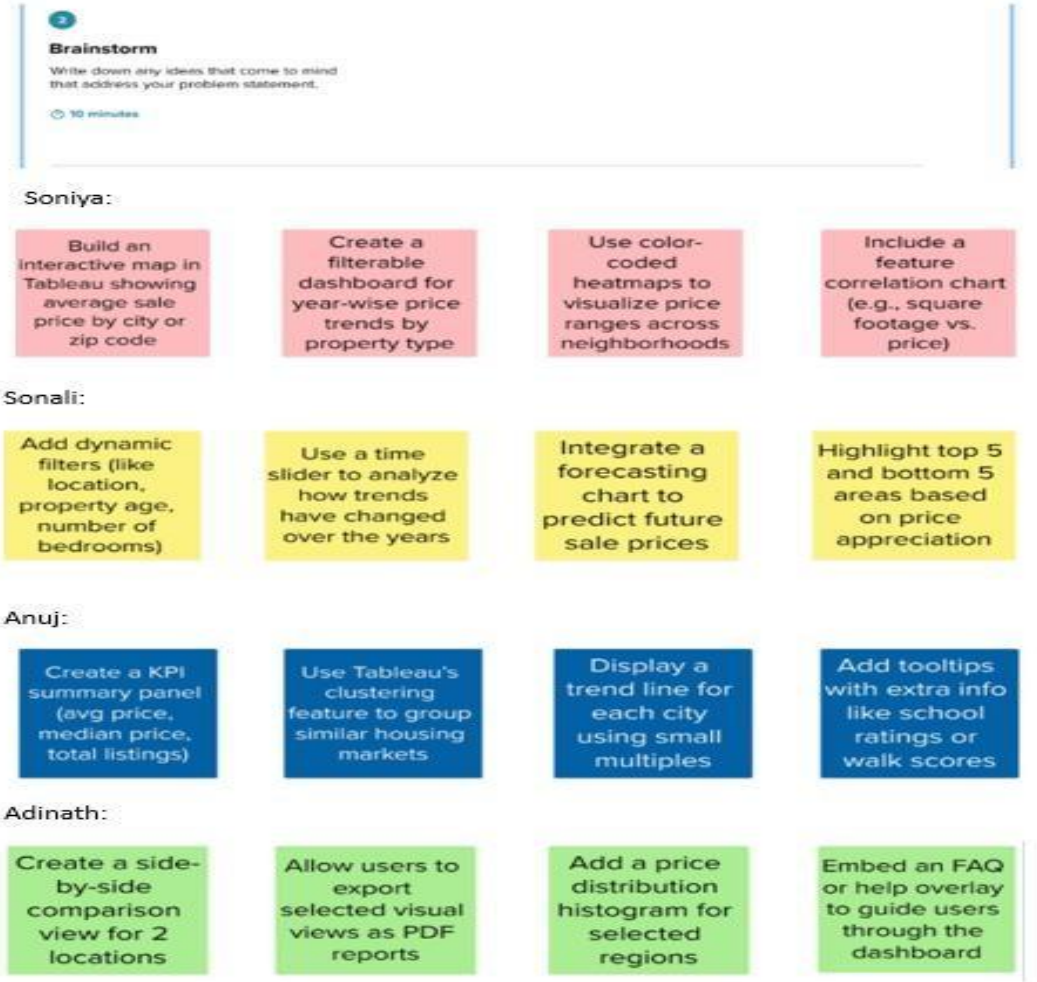
Problem ID	I am (Customer)	I'm trying to…	But…	Because…	Which makes me feel…

PS-1	A real estate analyst	Analyze housing market trends	The raw housing data is messy and multisource	Reports take too long to prepare and are inconsistent	Frustrated and overwhelmed
PS-2	A marketing team member	Identify trends in house features and sales	Data is incomplete or lacks proper structure	It is difficult to create visual reports for decisionmaking	Confused and stressed
PS-3	Company executive	Make pricing and investment decisions	Current dashboards do not highlight renovations, house age, or key features	Critical insights are hard to extract quickly	Anxious about missing key market opportunities

2.2 Empathy Map Canvas:

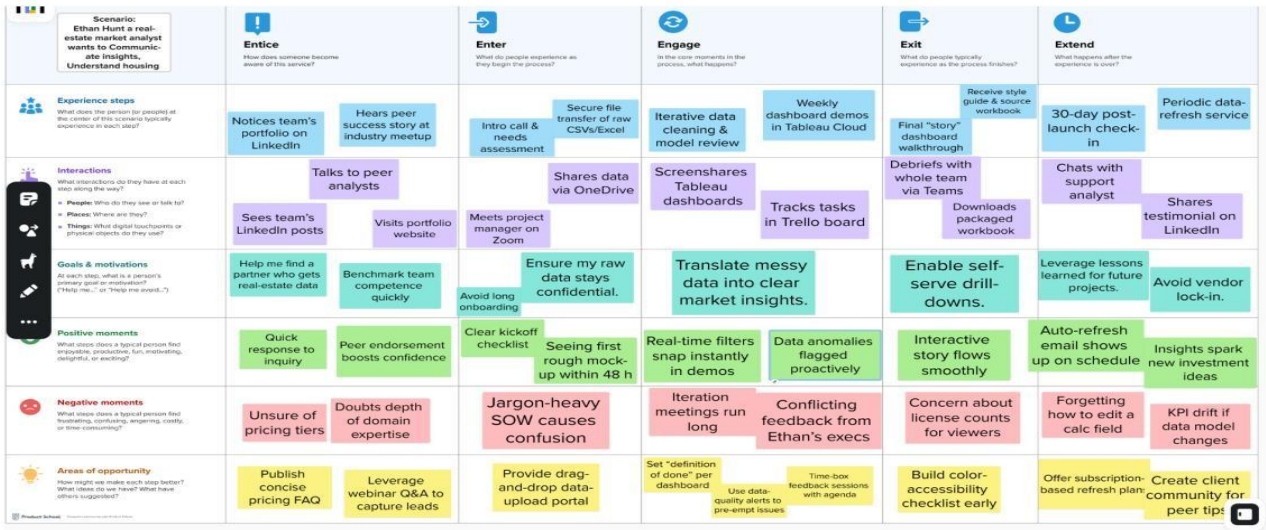


2.3 Brainstorming:



3) REQUIREMENT ANALYSIS:

3.1) Customer Journey map



3.2) Solution Requirement:

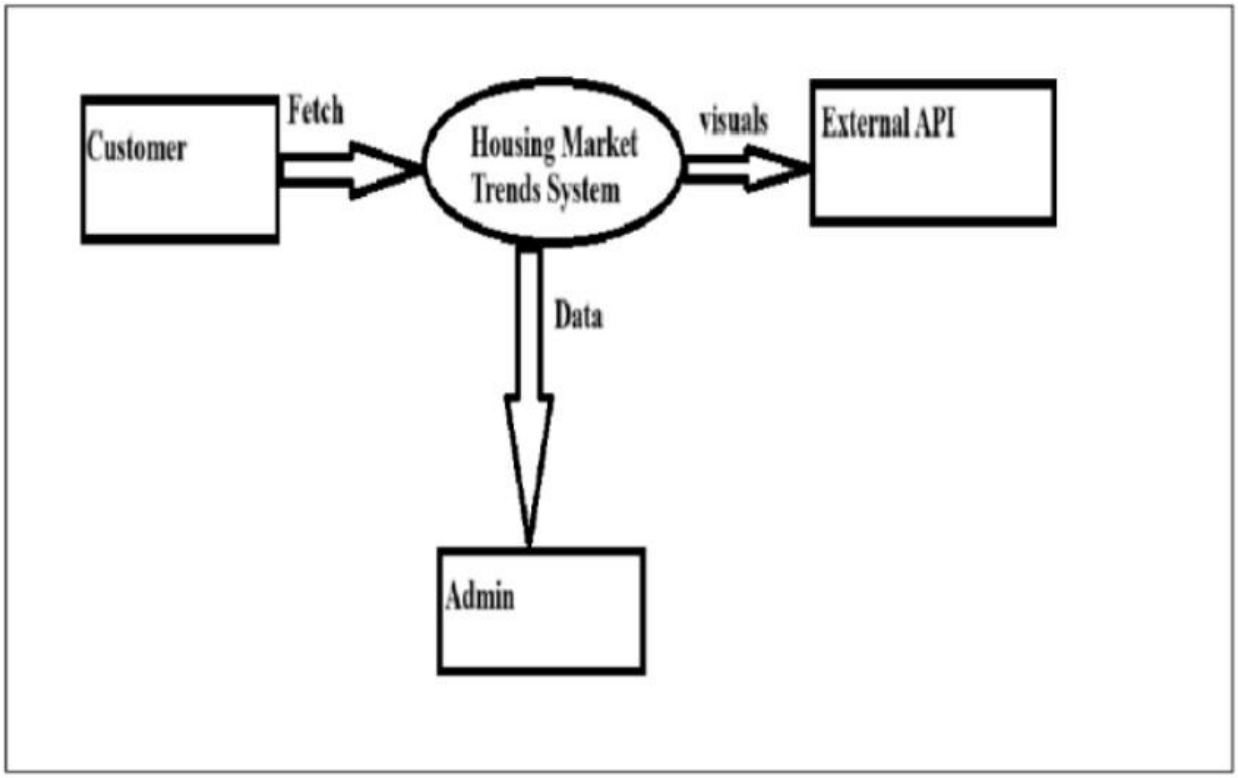
Functional Requirements: Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Upload Datasets	The system shall accept file formats such as .csv, .xlsx, and . json . The users shall be notified if the file format is invalid.
FR-2	Display Trends	The system shall provide line and area charts to show price trends. The system shall display tooltips with exact values on hover.
FR-3	Time Filtering	Users shall be able to select customer time intervales. The system shall provide quick filters like "Last 6 Months", "This Year"
FR-4	User Account Features	The system shall allow user registration and login. Logged-in users shall be able to save dashboards.

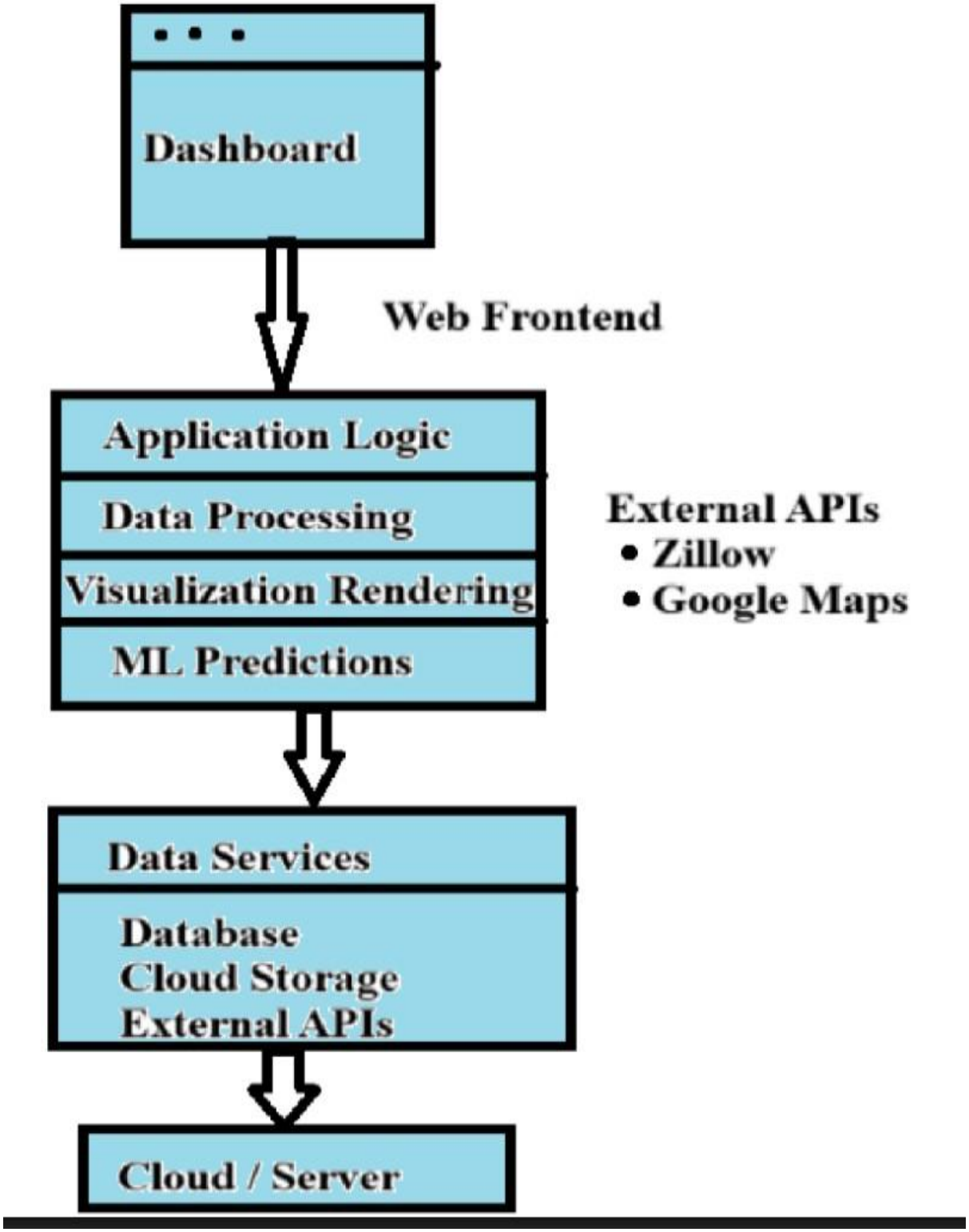
Non-functional Requirements: Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Focuses on how easy and intuitive the system is for users.
NFR-2	**Security**	Protects system data and operations from unauthorized access and misuse.
NFR-3	**Reliability**	Ensures the system performs correctly and consistently.
NFR-4	**Performance**	Refers to the speed and responsiveness of the system.
NFR-5	**Availability**	Focuses on ensuring the system is accessible and operational.
NFR-6	**Scalability**	Defines how the system adapts to increased workload or growth.

3.3) Data Flow Diagram:



3.4) Technical Stack:



4)PROJECT DESIGN:

4.1Problem Solution Fit:

Problem-Solution fit canvas 2.0

Purpose / Vision

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) — CS

Whoisyour customer?
i.e. working parents of 0-5 y.o. kids

Real-estate market analysts and strategists like Ethan Hunt, who work at property development firms, investment groups, or large brokerages.

6. CUSTOMER CONSTRAINTS — CC

Whatconstraints prevent yourcustomers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

Limited project budgets that require clear ROI justification. Tight deadlines dictated by business planning cycles.

5. AVAILABLE SOLUTIONS — AS

Which solutions areavailable tothecustomers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

Using Excel or basic built-in charting tools.

- Pros: No direct cost, total control.
- Cons: Extremely time-consuming, visuals are static and uninspiring, poor at handling geographic data, high risk of errors.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS — J&P

Which jobs-to-be-done(or problems)do youaddress for your customers? There could be more than one; explore different sides.

Translate raw, messy, and voluminous real-estate data (e.g., listings, sales, census data) into clear, interactive, and persuasive visualizations. Communicate market trends, opportunities, and risks to non-technical stakeholders (e.g., executives, investors) to drive strategic decisions.

9. PROBLEM ROOT CAUSE — RC

What is the real reason that thisproblem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

Word-of-mouth conversations with colleagues and industry peers. Attending real-estate industry conferences, seminars, and local meetups. Internal company meetings where the need for better analytics is discussed.

7. BEHAVIOUR — BE

Whatdoes yourcustomer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Direct Actions
Searching online for "real estate data visualization" or "Tableau developers for property analysis." Asking peers for recommendations. Attempting to clean the data themselves. Vetting vendor portfolios for relevant examples.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS — TR

What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

An upcoming quarterly business review, board meeting, or strategy session that requires a clear summary of market performance.

4. EMOTIONS: BEFORE / AFTER — EM

Howdo customers feelwhen theyface a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

Overwhelmed by data complexity. Frustrated with the limitations of existing tools. Anxious about deadlines and the clarity of their message. Uncertain about where the true insights are hidden in the raw numbers.

10. YOUR SOLUTION — SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

A dedicated data visualization service for real-estate professionals. We are a 4-person team that acts as an extension of the analyst's team, combining deep real-estate knowledge with expert-level Tableau skills.

8. CHANNELS of BEHAVIOUR — CH

8.1 ONLINE

What kind of actions do customers take online? Extract online channels from #7

Professional networks (Linkedin) for vetting experts and asking for referrals. Search engines (Google) for finding vendors. Vendor websites, blogs, and case studies. Video conferencing (Zoom/Teams) for discovery calls and demos. Secure cloud storage (OneDrive/Dropbox) for file sharing.

8.2 OFFLINE

What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

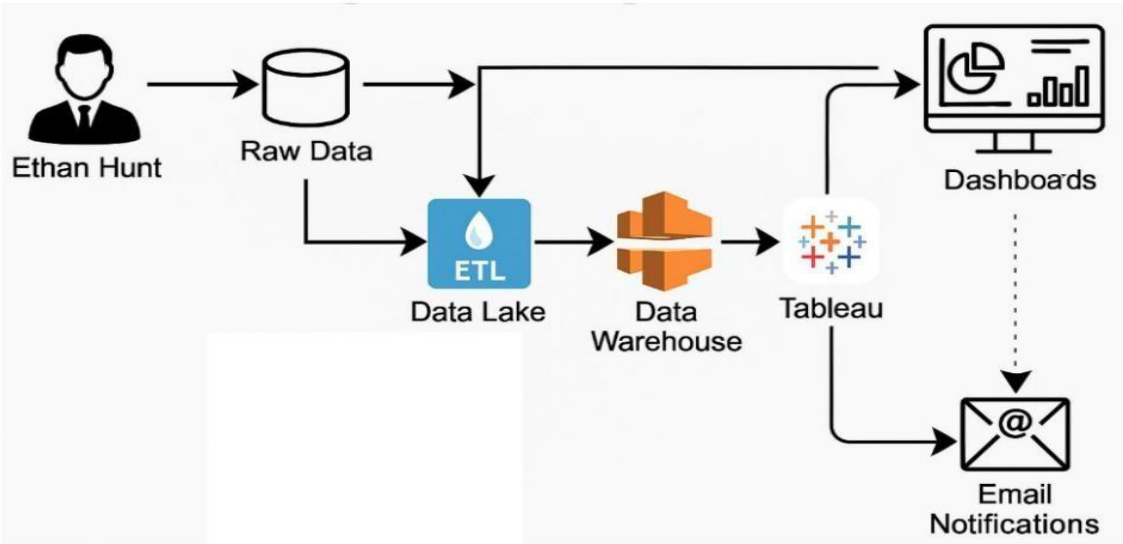
Word-of-mouth conversations with colleagues and industry peers. Attending real-estate industry conferences, seminars, and local meetups. Internal company meetings where the need for better analytics is discussed.

Extract online & offline CH of BE

4.2 Proposed Solution:

S. No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Real-estate analysts sit on vast, messy housing datasets but lack the time and tools to convert them into executive-ready insights.
2.	Idea / Solution description	Offer a turnkey Tableau service that cleans data and delivers interactive dashboards within 48 hours, followed by rapid collaborative iterations.
3.	Novelty / Uniqueness	Blends niche real-estate expertise with advanced Tableau skills, delivering geo-spatial heat maps and predictive overlays straight out of the box.
4.	Social Impact / Customer Satisfaction	Accelerates data-driven decisions that guide smarter urban development while cutting analyst stress and boosting stakeholder confidence.
5.	Business Model (Revenue Model)	Tiered, fixed-price projects plus optional monthly subscriptions for data refresh, support, and company-wide training upsells.
6.	Scalability of the Solution	Reusable ETL scripts and template dashboards let the team onboard more clients and expand into adjacent data domains without extra staffing.

4.3 Solution Architecture:



5) PROJECT PLANNING & SCHEDULING:

5.1 Project planning:

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	User Type
Sprint-1	Upload Housing Data	USN-1	As a user, I can upload CSV files containing housing market data.	2	High	Customer
Sprint-2	Fetch data from API	USN-2	As a user, I can fetch housing data from external APIs.	1	Medium	Customer
Sprint-2	Clean & Process Data	USN-3	As a user, I can apply filters & transformation to clean raw data.	2	High	Customer
Sprint-2	View Visualizations	USN-4	As a user, I can view housing market trends through interactive graphs and charts.	2	Medium	Customer
Sprint-3	Save & Export Reports	USN-5	As a user, I can download the generated reports in PDF/CSV format.	1	Low	Customer
Sprint-3	Manage Users	USN-6	As an admin, I can view and manage registered users.	2	High	Admin
Sprint-2	Dashboard Filtering	USN-7	As a user, I can filter the dashboard results by city, price range, or year	2	High	Customer
Sprint-4	View Usage Analytics	USN-8	As a admin , I can view how many users are using each feature.	2	High	Admin

6) FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

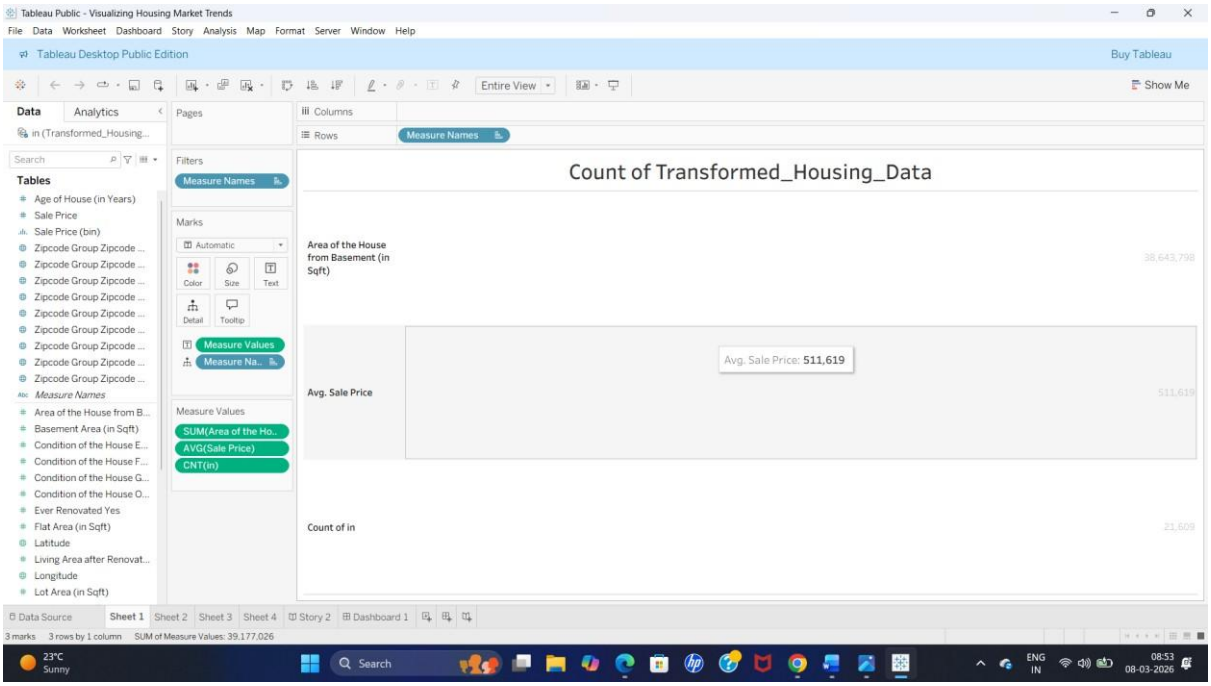
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Successfully imported and visualized cleaned housing market datasets (listings, prices, locations, sales trends) with no missing or corrupted values.
2.	Data Preprocessing	Processed raw files by removing duplicates, standardizing date and price formats, handling nulls, and joining supplemental datasets (e.g., census demographics).
3.	Utilization of Filters	Implemented interactive filters for region, price range, property type, listing status, and date, enabling users to dynamically explore market segments.
4.	Calculation fields Used	Created calculated fields for price per square foot, YoY % change, moving averages, and custom ranking scores to surface actionable trends.
5.	Dashboard design	Designed intuitive, multi-panel dashboards with color-coded trend lines, geospatial heatmaps, and KPI summary cards for rapid insights; responsive layouts for desktop/tablet.
6	Story Design	Developed an end-to-end Tableau Story with sequential sheets: Market Overview → Deep Dives → Opportunity Hotspots → Strategic Recommendations, guiding users through the insights narrative.

7)RESULTS:

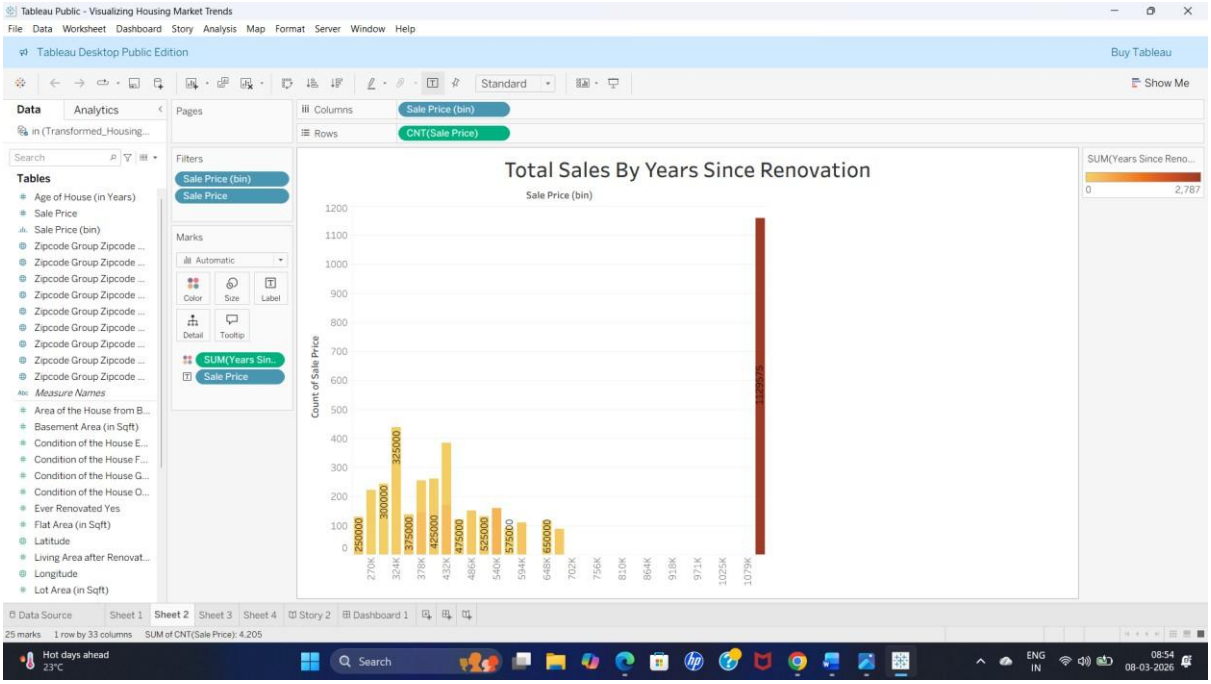
7.1 Output Screenshots:

Visualizations:

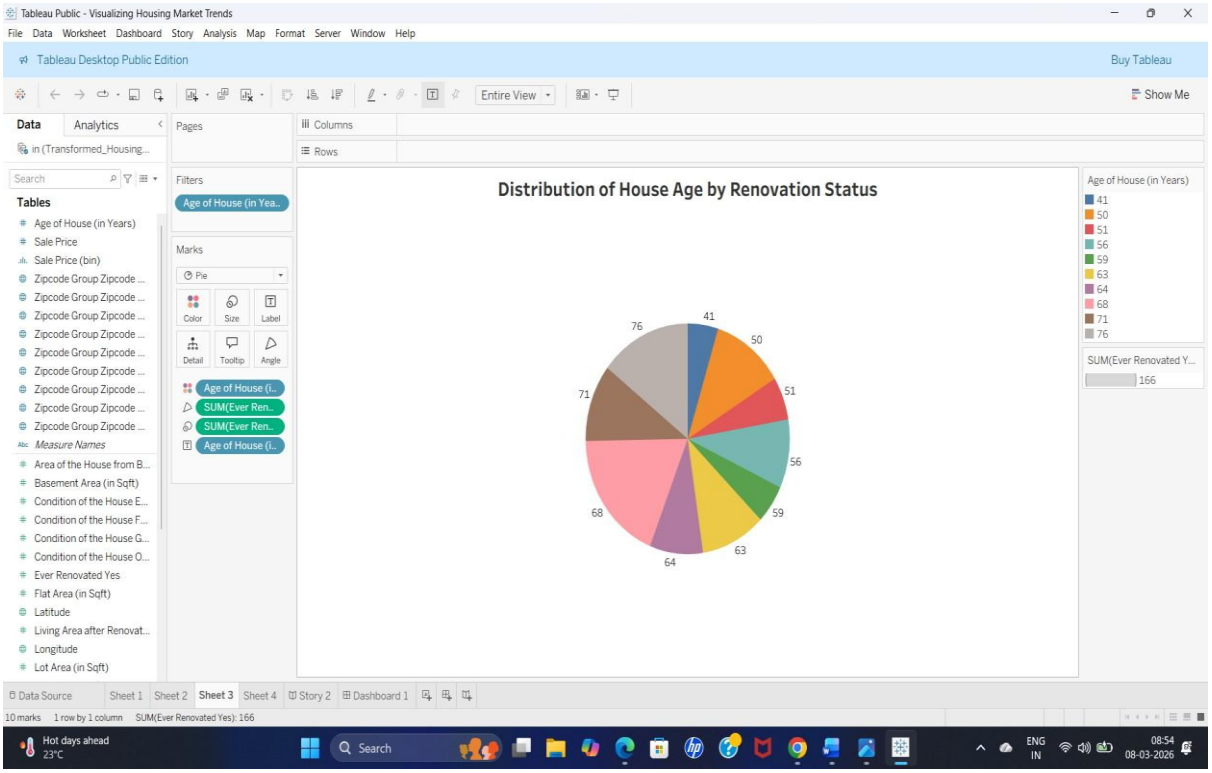
Sheet 1:



Sheet 2:



Sheet 3:



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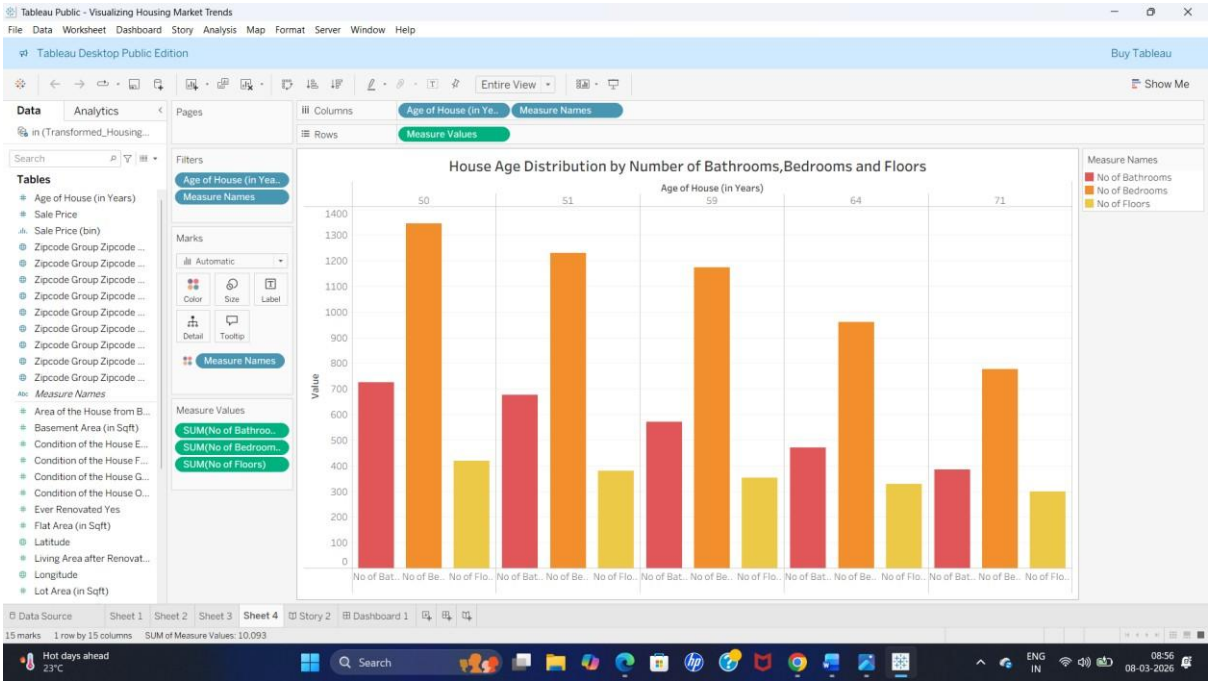
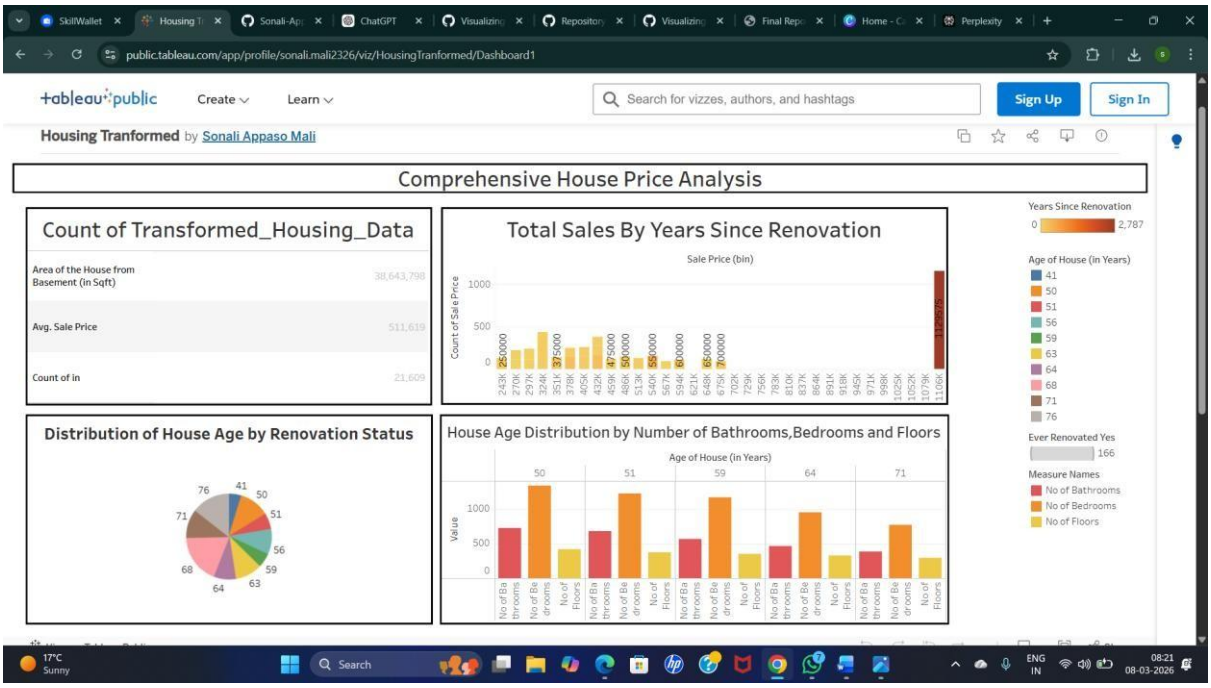
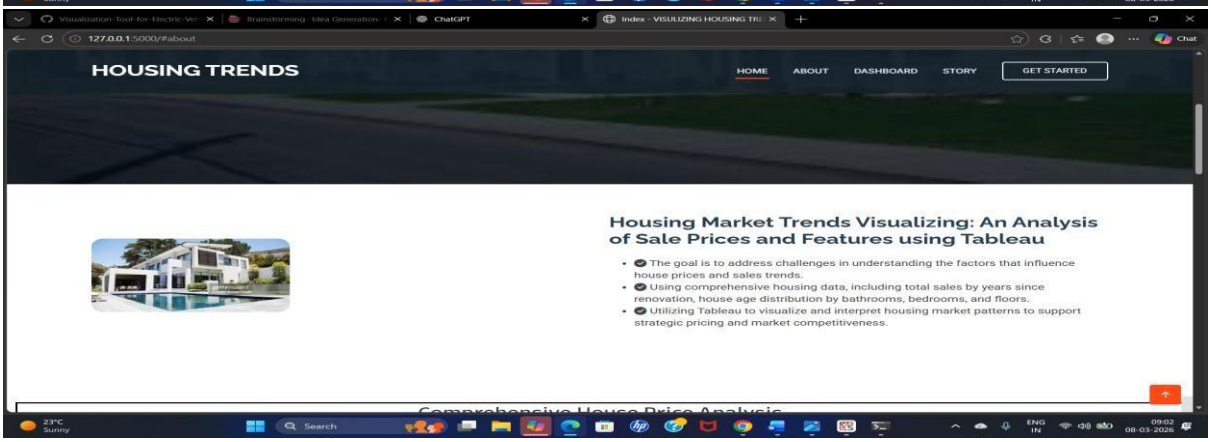


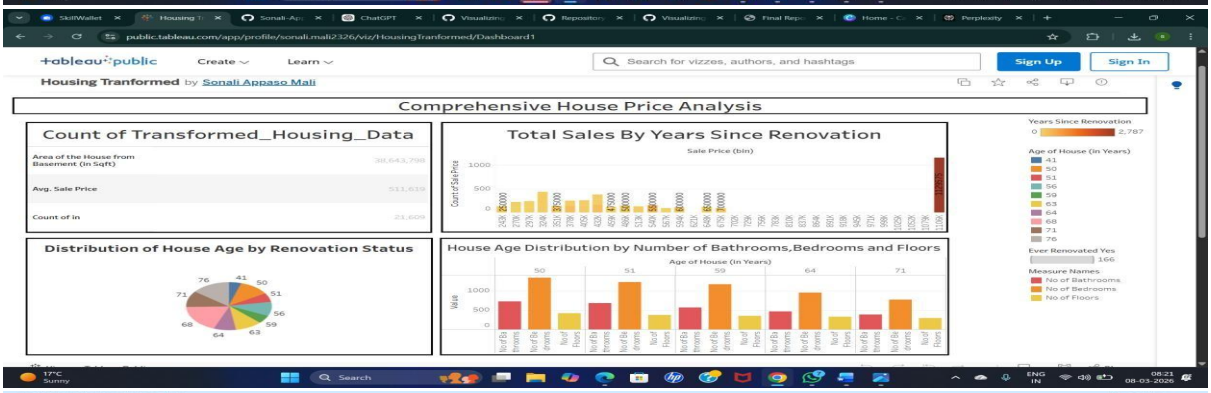
Dashboard:

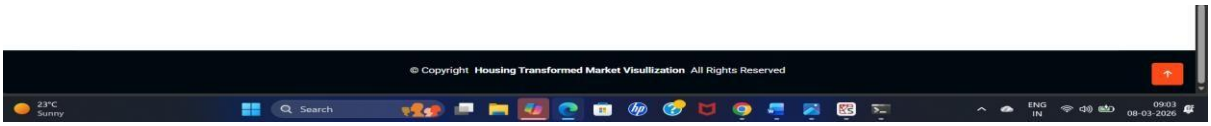


Story:

Web Integration with Spyder:







8) ADVANTAGES & DISADVANTAGES

Advantages

Rapid Insight Delivery

Our prototype-first workflow produces an initial Tableau dashboard within 48 h, letting Ethan validate direction early and keep the project tightly aligned with stakeholder needs. This shortens analysis cycles and builds executive confidence faster than traditional, sequential BI engagements.

Domain-Tailored Expertise

The four-person team combines real-estate knowledge with advanced Tableau and data-engineering skills, delivering geo-spatial heat maps, predictive overlays, and accessibility-ready colour palettes that generic consultants rarely offer. Ethan gains visuals that speak his market's nuances without needing to train an external vendor on the basics.

Disadvantages

Specialized Cost Structure

Premium, niche talent and rapid-turnaround service come at a higher price point than DIY approaches or commodity BI freelancers. For firms with tight budgets, the tiered fixed-fee plus optional subscription model may still feel prohibitive.

Vendor Dependency Risk

Although we provide training and reusable Prep flows, Ethan could remain reliant on our team for complex updates or new analytic modules. If internal capacity or Tableau proficiency doesn't grow in parallel, long-term dependence may limit flexibility and increase total cost of ownership.

9) CONCLUSION:

Our Tableau-driven solution converts Ethan's raw housing data into a clear, interactive narrative that accelerates insight and boosts executive confidence. By coupling rapid prototypes with reusable Prep flows, we've built a framework that delivers value today while remaining easy to refresh and scale tomorrow. With dashboards live and training complete, Ethan is now equipped to guide smarter real-estate decisions and continually refine his market intelligence,

10) FUTURE SCOPE:

Next, we'll plug in live MLS and economic feeds so the dashboards refresh in real-time and flag market anomalies automatically. The framework will be broadened to cover rental, commercial, and cross-city datasets, letting Ethan clone the solution to new regions with minimal re-work. We'll also embed predictive pricing models and auto-generated narrative insights, giving his team forward-looking guidance and instant executive summaries.

11) APPENDIX:

Dataset Link:

https://1drv.ms/x/c/fb9a1acd01532fe9/IQBy2HCb4LWvSI7YRaTLBkQbAYUhj32sURW8EE7_KNgEE9E?e=G5nZqt

Project Demo Link:

https://drive.google.com/file/d/1g6QC2HI7ekOu1gBj8b55UAXfvk83 -c8b/view?usp=drivesdk